Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/16/2007
PI Name:	Miller, David Ph.D.		
Project Title:	High Pressure Cool Flames and Auto-ignition at Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-2816	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (11/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Pearlman, Howard (Drexel University) Cernansky, Nicholas (Drexel University)		
Grant/Contract No.:	NNC04AA05A		
Performance Goal No.:			
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Task Description:	The original program involved a ground-based microgravity study on low-temperature reactions, cool flames and auto-ignition at elevated pressures (1atm P 10atm) is underway. This work is an extension of the reduced gravity, subatmospheric cool flame studies performed by Pearlman and co-workers (Pearlman, 1999, 2000). Extensions to elevated pressures will allow us to explore cool flames at conditions closer to the pressures of the critical applications in combustion systems. The program aims to isolate the roles of species and thermal diffusion on reaction front structure, stability, and evolution. On Earth, natural convection in unstirred, static reactors dominates diffusive transport and complicates the temperature and species concentration profiles. Moreover, these complexities cannot be effectively suppressed in terrestrial environments by reducing the pressure and/or reaction vessel size, i.e., Ra. At reduced gravity, however, buoyant convection can be effectively suppressed, the problem reduced from two (or three) dimensions to one-dimension, and modeled exactly. Develop ignition diagrams (phase plots of temperature versus pressure and mixture stoichiometry) for heavy hydrocarbon-air and hydrocarbon-oxygen mixtures at elevated pressures in the range of 1-10 atm. Measure and record the one-dimensional flame speed, integrated spatial distribution of uv and visible light, 2D formaldehyde and temperature distribution, and induction times associated with high pressure cool flames and autoignition fronts. Develop chemical kinetic models for high pressure, low-temperature reactions in a microgravity environment and compare the results to the experimental measurements. Compare the results to those obtained by other researcher efforts that have used CSTR's, heated flow tubes, and motored engines. As the program progressed it became necessary to refocus the work away from experimental studies to model development. A significant portion of the revised work plan involved developing chemical kinetic models of hydrocarbons which could be used to support the experimental work of a companion project of Dr. H. Pearlman. An improved understanding of the chemical basis for hydrocarbon ignition will improve the efficiency of combustion based power generation systems and the safety of fuel storage. This program is focused at pressures closer to those encountered in a combustion system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Ground based microgravity experiments will allow us to experimentally measure temperature, pressure, and selected species concentration profiles during the preignition reactivity of propane, butane, and heptane over a range of initial temperatures, pressures, and stoichiometries. This data will not include the effects of buoyancy, which complicate analysis and modeling and 1 g, and allow more complete determination of the important chemical and diffusive effects on ignition. An improved understanding of the chemical basis for hydrocarbon ignition will improve the efficiency of combustion based power generation systems and the safety of fuel storage. This program is focused at pressures closer to those encountered in a combustion system.
Task Progress:	During the first two years of the program the efforts were focused on developing a high pressure test rig with optical access for use in the ground based microgravity facilities. The primary issue with the design was selection of a non-reactive material or material coating that would withstand the targeted pressure range of the study and provide a non-catalytic surface. While our original design had proposed to use the pressurized quartz vessel-inside-a metal pressure vessel technique, which we have successfully utilized in our flow reactor facility, the possibility of rapid pressure rise inside the constant volume quartz reactor necessitated moving to an all metal reactor design which necessitated selection of an inert coating for the inner surface. Finding such a coating took much of our first eighteen months time as no manufacturer would guarantee the passivity and the durability of their products over the temperature (600-900 K) and pressure (1-10 atm) ranges of the program. It was during this phase of the preliminary design that we began to work on simulations of the oxidation processes that would occur within the reactor. These simulations identified the risk of the rapid pressure rise and also the need to run multi-dimensional calculations. Such calculations required us to take detailed kinetic models and develop lower dimensionality chemical models, also known as "reduced" models in order to perform a realistic simulation. About the time that we had identified possible coatings for our metal reactor to test their applicability, it became obvious that by the time we identified the appropriate coating, designed the facility, ran through the appropriate safety certification, then begin fabrication that there was going to be a significant reduction in the availability of time with the microgravity facilities. At this point we got permission to change the focus of the research program from primarily an experimental program to a modeling study of hydrocarbon oxidation. This work would also support the experimental work underway under the direction of a Co-Investigator of this program, Dr. H. Pearlman. We also undertook the development of optical diagnostics for in-situ measurements that were initially planned in collaboration with the ZARM. During the course of this work we have made significant progress, preparing five manuscripts for presentations which will ultimately be developed into archival publications and form the basis for dissertations. The work encompasses a range of models including detailed chemical formulations for propane, butane, n-heptane and iso-octane, the primary reference fuels (PRFs) for spark ignition engines, to explore the oxidation behavior of real size hydrocarbons, and reduced models for the propane and the PRFs to investigate mixtures, and global models of the PRFs to allow computations in two dimensions. The first manuscript described a global kinetic model that simulated the autoignition and ultimate combustion of propane at temperatures from 600 – 1000 K, for pressures ranging from 1 – 10 atm and over the range $0.5 < F < 4.0$. The reaction set included steps to model chain propagation, termination and branching reactions in the low temperature region, and a 2-step high temperature oxidation scheme. To ensure proper representation of NTC behavior, a reversible reaction or a self catalytic reaction was included. The model was validated against existing detailed mechanisms and published data in static and stirred reactors, and it was then compared to data from a spherical reactor at microgravity using 2-D computations. It reproduced the symmetrical reaction field of the spherical reactor. It was during the initial development of this model that we had identified the possible safety problems with the quartz vessel within a metal vessel design for our experiment. Later we compared a detailed kinetic model for propane with a skeletal model of 49 species and 80 reactions in 0D simulations in preparation for use in multidimensional calculations. The skeletal model was constructed entirely from

elementary reactions and rate parameters were selected from the literature. The skeletal mechanism predicted the quantitative global parameters (pressure, temperature, and induction times) and species concentrations, such as carbon monoxide, hydroxyl radical, formaldehyde, and cool flame light intensity very well, within the accuracy targets for the study. The mechanism was also able to predict the qualitative features including the number of cool flames as well as prediction of the transition to hot ignition. The mechanism is valid across a wide range of equivalence ratios. The skeletal mechanism reduced the computation time by a factor of three and also required an order of magnitude less computer memory for the computation than a detailed propane model. The mechanism will be used to predict the spatio-temporal temperature and species distribution in unstirred cool flame experiments using a two dimensional CFD model.

As mentioned, in keeping with our plan of studying more realistic fuels than propane, we have developed detailed and skeletal models of the PRFs for spark ignition engines. We have used these models to explore the effect of the ignition promoting additive di-tertiary butyl peroxide (DTBP) on their preignition chemistry, which direct application to homogeneous charge compression ignition engines. Both models were able to reproduce existing experimental data on preignition reactivity for a range of n-heptane/iso-octane ratios with a without the addition of up to 1.5% (by volume) DTBP. A key question to be answered in this study was to identify the mechanism by which DTBP promotes ignition. The decomposition of DTBP generates both moderate heat release and a pool of highly reactive radical intermediates and there is disagreement as to which mechanism promotes ignition. By examining the results from the simulation, we determined that for mixtures which were primarily more than 50% n-heptane the key mechanism for promotion is the additional heat release; however, for the mixtures that were primarily iso-octane the mechanism is the generation of the radical pool. This illustrates the power of combining experimental work with modeling studies. Both models led to similar interpretations; however, the skeletal model was computationally efficient enough (cutting computational time by two orders of magnitude; memory allocation by three orders of magnitude) that we are trying to apply it to multidimensional computations as well.

Unfortunately, our optical diagnostic work has not been as successful. We have been faced with an irritating sequence of equipment failures that have essentially kept this portion of the work from moving forward.

Bibliography Type:	Description: (Last Updated: 11/16/2007)
Papers from Meeting Proceedings	Gupta A, Miller DL, Cernansky NP. "Effect of DTBP on PRF Combustion in HCCI Engines." 2006 Technical Meeting of the Central States Section of the Combustion Institute, Cleveland, OH, May 2006. Paper CSSCI-43, Central States Section of the Combustion Institute, May 2006. , May-2006
Papers from Meeting Proceedings	Gupta A, Cernansky NP, Pearlman H, Miller DL. "A Numerical Study of the Influence of DTBP Addition on PRF Combustion in HCCI Engines." Presented at the 31st Symposium (Intl) of the Combustion Institute, Heidelberg, GDF, 10 August 2006. Poster WIPP 4E22, 31st Symposium (Intl) of the Combustion Institute, August 2006. Combustion Institute , Aug-2006
Papers from Meeting Proceedings	Gupta A, Foster M, Pearlman H, Miller DL, Cernansky NP. "Cool Flames at Microgravity: A Skeletal Kinetic Mechanism for Propane Oxidation." Presented at the 5th U.S. Combustion Meeting, San Diego, CA, March 2007. Paper USC C-32, 5th U.S. Combustion Meeting, Western States Section of the Combustion Institute, March 2007. , Mar-2007
Papers from Meeting Proceedings	Gupta A, Gong X, Miller DL, Cernansky NP. "A Detailed Kinetic Study on the Effect of DTBP on PRF Combustion in HCCI Engines." 2007 Power Train and Fluid Systems Conference, Kyoto, Japan, July 2007. SAE Paper No. 2007-01-2002 and JSAE 20077273, 2007. , Jul-2007
Papers from Meeting Proceedings	Gupta A, Miller DL, Pearlman H, Cernansky NP. "A Global Kinetic Model for Autoignition of Propane." Fourth Joint Meeting of the U.S. Section of the Combustion Institute, Philadelphia, PA, March 20-23, 2005. Paper JMUSSCI C11, Combustion Institute, 2005. , Mar-2005
Papers from Meeting Proceedings	Gupta A, Miller DL, Cernansky NP. "An Augmented Skeletal Mechanism for PRFs and Oxygenated Additives." Fifth U.S. Combustion Meeting, San Diego, CA, March 2007. Paper US E04, Fifth U.S. Combustion Meeting, Western States Section of the Combustion Institute, March 2007. , Mar-2007